

Warm-up 1

Give three different examples of matrices in $M_{2 \times 2}^{\mathbb{R}}$ that are in reduced row echelon form. In each one, circle each of the pivots.

Warm-up 2

What is the elementary matrix that, when applied to the given matrix on the left, carries out each of the following row moves?

$$\begin{pmatrix} 1 & 1 & 4 \\ 0 & 2 & 6 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & 4 \\ 0 & 1 & 3 \end{pmatrix}$$

$$\begin{pmatrix} 5 & 6 & 1 & 8 \\ 2 & 4 & 0 & 9 \\ 0 & 1 & 0 & 1 \\ 1 & 2 & 8 & 12 \\ 4 & 4 & 4 & 4 \end{pmatrix} \rightarrow \begin{pmatrix} 5 & 6 & 1 & 8 \\ 4 & 4 & 4 & 4 \\ 0 & 1 & 0 & 1 \\ 1 & 2 & 8 & 12 \\ 2 & 4 & 0 & 9 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 8 & 9 & 0 \\ -3 & 1 & 4 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 2 & 3 \\ 12 & 17 & 12 \\ -3 & 1 & 4 \end{pmatrix}$$

During class 1

Describe in vector form the solution set to the homogeneous system corresponding to the following matrix in RREF.

$$\begin{pmatrix} 1 & 0 & 8 \\ 0 & 1 & -3 \end{pmatrix}$$